Table 15 WHO base anthropometric outcomes for middle school, oldest child

Current school enrollment defines the middle school sample. One oldest child per household is selected within this level. WHO 2007 sex-specific monthly growth references use age on 16 December 2022. WAZ is defined only through month 120. Father and mother reform estimates use the established weighted RDD.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.124	-0.144 [0.219] N=1,398	-0.050 [0.217] N=1,253	-0.032 [0.232] N=1,075	-0.014 [0.263] N=890	-0.073 [0.298] N=712
Weight-for-age z score, through 120 months	0.484	0.999 [0.632] N=34	1.046 [0.667] N=32	**1.393* [0.780] N=28**	**1.616* [0.914] N=26**	0.880 [0.972] N=24
BMI-for-age z score	0.412	-0.034 [0.211] N=1,396	0.030 [0.220] N=1,252	-0.067 [0.235] N=1,074	0.030 [0.231] N=890	-0.026 [0.254] N=713
Stunted, HAZ below -2	0.106 P=147	0.046 [0.047] N=1,398	0.046 [0.049] N=1,253	0.026 [0.053] N=1,075	0.061 [0.060] N=890	0.105 [0.065] N=712
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.060 P=80	-0.030 [0.035] N=1,396	-0.037 [0.038] N=1,252	-0.011 [0.029] N=1,074	-0.013 [0.032] N=890	0.012 [0.033] N=713
Overweight including obesity, BMIZ above +1	0.381 P=533	-0.018 [0.082] N=1,396	0.014 [0.083] N=1,252	-0.000 [0.091] N=1,074	0.032 [0.094] N=890	0.017 [0.107] N=713
Obesity, BMIZ above +2	0.116 P=165	-0.012 [0.043] N=1,396	-0.014 [0.045] N=1,252	-0.039 [0.050] N=1,074	-0.016 [0.043] N=890	-0.013 [0.048] N=713

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.388 [0.337] N=1,227	-0.226 [0.324] N=1,082	-0.236 [0.368] N=904	-0.157 [0.429] N=719	-0.436 [0.584] N=541
Weight-for-age z score, through 120 months	0.434 [1.174] N=28	0.499 [1.222] N=26	-0.675 [1.455] N=22	-0.017 [1.866] N=20	-1.415 [1.832] N=18
BMI-for-age z score	-0.341 [0.366] N=1,225	-0.223 [0.400] N=1,081	-0.492 [0.404] N=903	-0.251 [0.443] N=719	-0.548 [0.611] N=542
Stunted, HAZ below -2	-0.009 [0.069] N=1,227	-0.019 [0.073] N=1,082	-0.062 [0.083] N=904	-0.016 [0.103] N=719	0.107 [0.129] N=541
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.025 [0.065] N=1,225	-0.046 [0.073] N=1,081	0.016 [0.052] N=903	0.002 [0.064] N=719	0.101 [0.069] N=542
Overweight including obesity, BMIZ above +1	-0.107 [0.127] N=1,225	-0.048 [0.127] N=1,081	-0.093 [0.139] N=903	-0.005 [0.157] N=719	-0.086 [0.217] N=542
Obesity, BMIZ above +2	-0.040 [0.076] N=1,225	-0.048 [0.082] N=1,081	-0.088 [0.093] N=903	-0.036 [0.082] N=719	0.007 [0.112] N=542

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.124	-0.145 [0.219] N=1,398	-0.054 [0.219] N=1,253	-0.031 [0.234] N=1,075	-0.015 [0.265] N=890	-0.068 [0.301] N=712
Weight-for-age z score, through 120 months	0.484	0.776 [0.664] N=34	0.772 [0.711] N=32	**1.250* [0.694] N=28**	1.373 [0.866] N=26	0.864 [0.917] N=24
BMI-for-age z score	0.412	-0.035 [0.203] N=1,396	0.020 [0.213] N=1,252	-0.062 [0.231] N=1,074	0.027 [0.225] N=890	-0.004 [0.247] N=713
Stunted, HAZ below -2	0.106 P=147	0.046 [0.047] N=1,398	0.046 [0.049] N=1,253	0.026 [0.053] N=1,075	0.061 [0.061] N=890	0.103 [0.065] N=712
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.060 P=80	-0.030 [0.035] N=1,396	-0.037 [0.037] N=1,252	-0.011 [0.029] N=1,074	-0.013 [0.032] N=890	0.010 [0.033] N=713
Overweight including obesity, BMIZ above +1	0.381 P=533	-0.019 [0.081] N=1,396	0.010 [0.082] N=1,252	0.001 [0.090] N=1,074	0.031 [0.093] N=890	0.025 [0.107] N=713
Obesity, BMIZ above +2	0.116 P=165	-0.012 [0.042] N=1,396	-0.015 [0.044] N=1,252	-0.038 [0.049] N=1,074	-0.016 [0.042] N=890	-0.011 [0.047] N=713

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.394 [0.331] N=1,227	-0.241 [0.323] N=1,082	-0.241 [0.367] N=904	-0.174 [0.428] N=719	-0.440 [0.587] N=541
Weight-for-age z score, through 120 months	0.356 [1.034] N=28	0.331 [1.084] N=26	-0.054 [1.497] N=22	0.022 [1.998] N=20	-1.114 [1.962] N=18
BMI-for-age z score	-0.351 [0.357] N=1,225	-0.272 [0.391] N=1,081	-0.509 [0.397] N=903	-0.312 [0.438] N=719	-0.606 [0.600] N=542
Stunted, HAZ below -2	-0.009 [0.069] N=1,227	-0.019 [0.073] N=1,082	-0.062 [0.083] N=904	-0.015 [0.103] N=719	0.109 [0.130] N=541
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.025 [0.065] N=1,225	-0.042 [0.072] N=1,081	0.017 [0.052] N=903	0.007 [0.064] N=719	0.107 [0.070] N=542
Overweight including obesity, BMIZ above +1	-0.110 [0.124] N=1,225	-0.065 [0.126] N=1,081	-0.099 [0.137] N=903	-0.028 [0.156] N=719	-0.107 [0.218] N=542
Obesity, BMIZ above +2	-0.041 [0.077] N=1,225	-0.053 [0.082] N=1,081	-0.089 [0.093] N=903	-0.040 [0.082] N=719	0.004 [0.113] N=542

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.124	-0.118 [0.221] N=1,398	-0.024 [0.217] N=1,253	-0.025 [0.226] N=1,075	-0.029 [0.261] N=890	-0.080 [0.296] N=712
Weight-for-age z score, through 120 months	0.484	0.817 [0.814] N=34	0.840 [0.812] N=32	0.868 [0.645] N=28	1.133 [0.824] N=26	0.496 [0.869] N=24
BMI-for-age z score	0.412	-0.033 [0.199] N=1,396	0.025 [0.209] N=1,252	-0.066 [0.226] N=1,074	0.015 [0.220] N=890	-0.013 [0.244] N=713
Stunted, HAZ below -2	0.106 P=147	0.043 [0.048] N=1,398	0.044 [0.051] N=1,253	0.026 [0.054] N=1,075	0.063 [0.062] N=890	0.108 [0.066] N=712
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.060 P=80	-0.028 [0.035] N=1,396	-0.036 [0.038] N=1,252	-0.010 [0.029] N=1,074	-0.011 [0.033] N=890	0.007 [0.034] N=713
Overweight including obesity, BMIZ above +1	0.381 P=533	-0.015 [0.079] N=1,396	0.015 [0.080] N=1,252	0.003 [0.088] N=1,074	0.027 [0.091] N=890	0.023 [0.105] N=713
Obesity, BMIZ above +2	0.116 P=165	-0.011 [0.042] N=1,396	-0.015 [0.044] N=1,252	-0.039 [0.049] N=1,074	-0.019 [0.044] N=890	-0.018 [0.050] N=713

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.382 [0.331] N=1,227	-0.222 [0.316] N=1,082	-0.291 [0.350] N=904	-0.276 [0.433] N=719	-0.533 [0.584] N=541
Weight-for-age z score, through 120 months	0.589 [1.391] N=28	0.620 [1.395] N=26	-0.004 [2.217] N=22	-0.085 [2.665] N=20	-1.334 [2.864] N=18
BMI-for-age z score	-0.346 [0.354] N=1,225	-0.258 [0.388] N=1,081	-0.544 [0.398] N=903	-0.340 [0.450] N=719	-0.666 [0.624] N=542
Stunted, HAZ below -2	-0.018 [0.071] N=1,227	-0.030 [0.075] N=1,082	-0.067 [0.084] N=904	-0.018 [0.107] N=719	0.095 [0.134] N=541
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.019 [0.066] N=1,225	-0.038 [0.074] N=1,081	0.030 [0.052] N=903	0.020 [0.065] N=719	0.117 [0.071] N=542
Overweight including obesity, BMIZ above +1	-0.106 [0.120] N=1,225	-0.063 [0.121] N=1,081	-0.109 [0.133] N=903	-0.042 [0.154] N=719	-0.117 [0.218] N=542
Obesity, BMIZ above +2	-0.039 [0.076] N=1,225	-0.052 [0.082] N=1,081	-0.086 [0.094] N=903	-0.035 [0.082] N=719	0.008 [0.112] N=542

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.124	-0.174 [0.216] N=1,398	-0.077 [0.211] N=1,253	-0.074 [0.224] N=1,075	-0.068 [0.259] N=890	-0.130 [0.291] N=712
Weight-for-age z score, through 120 months	0.484	-0.173 [0.800] N=34	-0.103 [0.823] N=32	0.884 [0.754] N=28	1.116 [0.951] N=26	0.431 [1.291] N=24
BMI-for-age z score	0.412	-0.045 [0.203] N=1,396	0.011 [0.210] N=1,252	-0.074 [0.226] N=1,074	0.005 [0.221] N=890	-0.018 [0.246] N=713
Stunted, HAZ below -2	0.106 P=147	0.049 [0.047] N=1,398	0.050 [0.049] N=1,253	0.032 [0.054] N=1,075	0.067 [0.061] N=890	**0.110* [0.065] N=712**
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.060 P=80	-0.028 [0.036] N=1,396	-0.034 [0.038] N=1,252	-0.009 [0.029] N=1,074	-0.011 [0.032] N=890	0.010 [0.034] N=713
Overweight including obesity, BMIZ above +1	0.381 P=533	-0.015 [0.080] N=1,396	0.016 [0.081] N=1,252	0.005 [0.090] N=1,074	0.027 [0.093] N=890	0.023 [0.108] N=713
Obesity, BMIZ above +2	0.116 P=165	-0.012 [0.042] N=1,396	-0.014 [0.044] N=1,252	-0.039 [0.048] N=1,074	-0.020 [0.042] N=890	-0.016 [0.047] N=713

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.429 [0.335] N=1,227	-0.243 [0.310] N=1,082	-0.267 [0.341] N=904	-0.253 [0.414] N=719	-0.486 [0.560] N=541
Weight-for-age z score, through 120 months	-0.889 [1.523] N=28	-0.403 [1.405] N=26	-0.907 [2.986] N=22	-0.089 [3.849] N=20	-1.405 [3.609] N=18
BMI-for-age z score	-0.342 [0.356] N=1,225	-0.252 [0.383] N=1,081	-0.501 [0.390] N=903	-0.328 [0.438] N=719	-0.624 [0.605] N=542
Stunted, HAZ below -2	-0.006 [0.071] N=1,227	-0.019 [0.074] N=1,082	-0.059 [0.083] N=904	-0.006 [0.105] N=719	0.109 [0.133] N=541
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.024 [0.066] N=1,225	-0.044 [0.073] N=1,081	0.016 [0.053] N=903	0.004 [0.066] N=719	0.104 [0.072] N=542
Overweight including obesity, BMIZ above +1	-0.106 [0.123] N=1,225	-0.057 [0.123] N=1,081	-0.097 [0.135] N=903	-0.046 [0.156] N=719	-0.129 [0.219] N=542
Obesity, BMIZ above +2	-0.042 [0.075] N=1,225	-0.051 [0.081] N=1,081	-0.088 [0.092] N=903	-0.049 [0.083] N=719	-0.005 [0.113] N=542

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.133	-0.097 [0.216] N=1,328	0.010 [0.214] N=1,190	0.038 [0.238] N=1,021	0.074 [0.277] N=847	-0.005 [0.323] N=677
Weight-for-age z score, through 120 months	0.466	**1.485* [0.819] N=33**	**1.561* [0.821] N=31**	**1.699* [0.893] N=27**	1.794 [1.210] N=25	1.177 [1.114] N=23
BMI-for-age z score	0.418	-0.116 [0.213] N=1,327	-0.062 [0.225] N=1,189	-0.133 [0.244] N=1,020	-0.054 [0.239] N=847	-0.075 [0.267] N=678
Stunted, HAZ below -2	0.107 P=143	0.019 [0.051] N=1,328	0.017 [0.054] N=1,190	-0.004 [0.059] N=1,021	0.030 [0.067] N=847	0.075 [0.072] N=677
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.059 P=75	-0.030 [0.037] N=1,327	-0.037 [0.040] N=1,189	-0.014 [0.031] N=1,020	-0.013 [0.035] N=847	0.004 [0.037] N=678
Overweight including obesity, BMIZ above +1	0.383 P=510	-0.046 [0.084] N=1,327	-0.020 [0.085] N=1,189	-0.029 [0.096] N=1,020	-0.000 [0.098] N=847	-0.013 [0.113] N=678
Obesity, BMIZ above +2	0.116 P=157	-0.031 [0.043] N=1,327	-0.038 [0.045] N=1,189	-0.060 [0.051] N=1,020	-0.036 [0.043] N=847	-0.038 [0.049] N=678

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.352 [0.307] N=1,168	-0.185 [0.291] N=1,030	-0.145 [0.344] N=861	-0.037 [0.425] N=687	-0.391 [0.600] N=517
Weight-for-age z score, through 120 months	0.663 [0.859] N=28	0.926 [0.928] N=26	0.739 [1.774] N=22	0.619 [2.277] N=20	-0.592 [1.997] N=18
BMI-for-age z score	-0.393 [0.343] N=1,167	-0.323 [0.379] N=1,029	-0.563 [0.371] N=860	-0.356 [0.405] N=687	-0.523 [0.555] N=518
Stunted, HAZ below -2	-0.024 [0.072] N=1,168	-0.037 [0.077] N=1,030	-0.089 [0.086] N=861	-0.045 [0.110] N=687	0.088 [0.136] N=517
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.017 [0.066] N=1,167	-0.034 [0.074] N=1,029	0.022 [0.054] N=860	0.014 [0.068] N=687	0.099 [0.073] N=518
Overweight including obesity, BMIZ above +1	-0.116 [0.120] N=1,167	-0.077 [0.121] N=1,029	-0.117 [0.133] N=860	-0.043 [0.151] N=687	-0.103 [0.215] N=518
Obesity, BMIZ above +2	-0.040 [0.077] N=1,167	-0.057 [0.082] N=1,029	-0.091 [0.094] N=860	-0.026 [0.082] N=687	0.020 [0.110] N=518

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.133	-0.105 [0.216] N=1,328	-0.006 [0.214] N=1,190	0.025 [0.238] N=1,021	0.060 [0.275] N=847	-0.021 [0.317] N=677
Weight-for-age z score, through 120 months	0.466	**1.489* [0.822] N=33**	1.438 [0.848] N=31	**1.699* [0.893] N=27**	1.794 [1.210] N=25	1.177 [1.114] N=23
BMI-for-age z score	0.418	-0.110 [0.212] N=1,327	-0.052 [0.224] N=1,189	-0.129 [0.243] N=1,020	-0.046 [0.237] N=847	-0.066 [0.263] N=678
Stunted, HAZ below -2	0.107 P=143	0.020 [0.051] N=1,328	0.019 [0.054] N=1,190	-0.004 [0.059] N=1,021	0.031 [0.066] N=847	0.075 [0.072] N=677
Underweight, WAZ below -2, through 120 months	0.000 P=0	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	0.059 P=75	-0.029 [0.037] N=1,327	-0.037 [0.040] N=1,189	-0.013 [0.031] N=1,020	-0.013 [0.035] N=847	0.004 [0.036] N=678
Overweight including obesity, BMIZ above +1	0.383 P=510	-0.040 [0.085] N=1,327	-0.011 [0.086] N=1,189	-0.021 [0.097] N=1,020	0.008 [0.099] N=847	-0.003 [0.114] N=678
Obesity, BMIZ above +2	0.116 P=157	-0.028 [0.043] N=1,327	-0.035 [0.045] N=1,189	-0.058 [0.050] N=1,020	-0.033 [0.042] N=847	-0.033 [0.048] N=678

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.339 [0.314] N=1,168	-0.178 [0.302] N=1,030	-0.121 [0.358] N=861	-0.007 [0.431] N=687	-0.362 [0.605] N=517
Weight-for-age z score, through 120 months	0.583 [0.810] N=28	0.413 [0.849] N=26	0.739 [1.774] N=22	0.619 [2.277] N=20	-0.592 [1.997] N=18
BMI-for-age z score	-0.394 [0.341] N=1,167	-0.322 [0.376] N=1,029	-0.563 [0.373] N=860	-0.368 [0.405] N=687	-0.544 [0.554] N=518
Stunted, HAZ below -2	-0.025 [0.073] N=1,168	-0.037 [0.079] N=1,030	-0.094 [0.088] N=861	-0.049 [0.111] N=687	0.082 [0.141] N=517
Underweight, WAZ below -2, through 120 months	NA	NA	NA	NA	NA
Thinness, BMIZ below -2	-0.017 [0.066] N=1,167	-0.033 [0.074] N=1,029	0.022 [0.054] N=860	0.015 [0.068] N=687	0.099 [0.073] N=518
Overweight including obesity, BMIZ above +1	-0.119 [0.122] N=1,167	-0.076 [0.123] N=1,029	-0.117 [0.134] N=860	-0.047 [0.151] N=687	-0.107 [0.215] N=518
Obesity, BMIZ above +2	-0.040 [0.077] N=1,167	-0.057 [0.082] N=1,029	-0.091 [0.095] N=860	-0.029 [0.082] N=687	0.015 [0.111] N=518

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.068	-0.117 [0.129] N=2,025	-0.204 [0.138] N=1,878	-0.201 [0.141] N=1,671	-0.172 [0.151] N=1,459	-0.164 [0.160] N=1,200
Weight-for-age z score, through 120 months	0.339	0.535 [0.974] N=43	0.999 [1.075] N=40	0.655 [0.963] N=38	0.879 [1.128] N=33	1.241 [1.175] N=29
BMI-for-age z score	0.425	-0.061 [0.153] N=2,021	-0.017 [0.160] N=1,875	-0.140 [0.173] N=1,667	-0.118 [0.187] N=1,455	-0.146 [0.207] N=1,199
Stunted, HAZ below -2	0.099 P=200	0.003 [0.027] N=2,025	0.003 [0.029] N=1,878	0.007 [0.029] N=1,671	0.018 [0.031] N=1,459	0.019 [0.034] N=1,200
Underweight, WAZ below -2, through 120 months	0.028 P=1	-0.148 [0.161] N=43	-0.122 [0.137] N=40	-0.105 [0.121] N=38	-0.156 [0.168] N=33	-0.093 [0.139] N=29
Thinness, BMIZ below -2	0.059 P=115	0.004 [0.024] N=2,021	-0.001 [0.026] N=1,875	0.014 [0.028] N=1,667	0.001 [0.028] N=1,455	-0.004 [0.030] N=1,199
Overweight including obesity, BMIZ above +1	0.378 P=773	0.035 [0.050] N=2,021	0.050 [0.053] N=1,875	0.029 [0.057] N=1,667	0.026 [0.062] N=1,455	0.028 [0.072] N=1,199
Obesity, BMIZ above +2	0.126 P=258	-0.051 [0.032] N=2,021	-0.047 [0.033] N=1,875	**-0.077** [0.035] N=1,667**	**-0.077** [0.039] N=1,455**	**-0.106** [0.042] N=1,199**

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.233 [0.196] N=1,697	**-0.411* [0.221] N=1,550**	**-0.488** [0.235] N=1,343**	**-0.485* [0.283] N=1,131**	**-0.661* [0.336] N=872**
Weight-for-age z score, through 120 months	-0.061 [1.457] N=34	0.939 [1.739] N=31	0.049 [1.906] N=29	2.489 [3.284] N=24	2.526 [3.778] N=20
BMI-for-age z score	-0.108 [0.219] N=1,694	-0.015 [0.241] N=1,548	-0.243 [0.275] N=1,340	-0.223 [0.324] N=1,128	-0.323 [0.403] N=872
Stunted, HAZ below -2	0.020 [0.044] N=1,697	0.017 [0.050] N=1,550	0.031 [0.050] N=1,343	0.051 [0.060] N=1,131	0.090 [0.073] N=872
Underweight, WAZ below -2, through 120 months	-0.192 [0.240] N=34	-0.089 [0.190] N=31	-0.010 [0.180] N=29	-0.021 [0.404] N=24	0.198 [0.398] N=20
Thinness, BMIZ below -2	0.051 [0.041] N=1,694	0.046 [0.046] N=1,548	0.084 [0.053] N=1,340	0.073 [0.058] N=1,128	0.094 [0.076] N=872
Overweight including obesity, BMIZ above +1	0.065 [0.063] N=1,694	0.102 [0.069] N=1,548	0.079 [0.078] N=1,340	0.084 [0.093] N=1,128	0.124 [0.127] N=872
Obesity, BMIZ above +2	0.007 [0.043] N=1,694	0.025 [0.047] N=1,548	-0.018 [0.053] N=1,340	0.006 [0.063] N=1,128	-0.039 [0.079] N=872

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.068	-0.111 [0.128] N=2,025	-0.193 [0.137] N=1,878	-0.187 [0.141] N=1,671	-0.155 [0.150] N=1,459	-0.152 [0.159] N=1,200
Weight-for-age z score, through 120 months	0.339	0.133 [1.155] N=43	0.567 [1.175] N=40	0.321 [1.065] N=38	0.769 [1.223] N=33	1.146 [1.325] N=29
BMI-for-age z score	0.425	-0.039 [0.149] N=2,021	0.018 [0.155] N=1,875	-0.108 [0.168] N=1,667	-0.079 [0.181] N=1,455	-0.110 [0.196] N=1,199
Stunted, HAZ below -2	0.099 P=200	0.002 [0.027] N=2,025	0.002 [0.029] N=1,878	0.006 [0.028] N=1,671	0.017 [0.031] N=1,459	0.019 [0.033] N=1,200
Underweight, WAZ below -2, through 120 months	0.028 P=1	-0.201 [0.203] N=43	-0.180 [0.181] N=40	-0.161 [0.163] N=38	-0.168 [0.179] N=33	-0.095 [0.140] N=29
Thinness, BMIZ below -2	0.059 P=115	0.003 [0.024] N=2,021	-0.002 [0.025] N=1,875	0.013 [0.028] N=1,667	-0.000 [0.028] N=1,455	-0.005 [0.030] N=1,199
Overweight including obesity, BMIZ above +1	0.378 P=773	0.043 [0.048] N=2,021	0.063 [0.051] N=1,875	0.041 [0.055] N=1,667	0.041 [0.060] N=1,455	0.041 [0.069] N=1,199
Obesity, BMIZ above +2	0.126 P=258	-0.047 [0.032] N=2,021	-0.041 [0.033] N=1,875	**-0.072** [0.035] N=1,667**	**-0.070* [0.039] N=1,455**	**-0.100** [0.042] N=1,199**

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.227 [0.194] N=1,697	**-0.395* [0.217] N=1,550**	**-0.469** [0.231] N=1,343**	-0.453 [0.276] N=1,131	**-0.631* [0.329] N=872**
Weight-for-age z score, through 120 months	-0.586 [1.834] N=34	0.529 [2.192] N=31	-0.354 [2.151] N=29	2.157 [3.459] N=24	1.920 [4.599] N=20
BMI-for-age z score	-0.088 [0.221] N=1,694	0.034 [0.242] N=1,548	-0.200 [0.276] N=1,340	-0.157 [0.327] N=1,128	-0.251 [0.404] N=872
Stunted, HAZ below -2	0.020 [0.044] N=1,697	0.016 [0.050] N=1,550	0.030 [0.050] N=1,343	0.049 [0.060] N=1,131	0.088 [0.073] N=872
Underweight, WAZ below -2, through 120 months	-0.290 [0.309] N=34	-0.264 [0.296] N=31	-0.184 [0.266] N=29	-0.128 [0.430] N=24	0.172 [0.451] N=20
Thinness, BMIZ below -2	0.050 [0.041] N=1,694	0.045 [0.045] N=1,548	0.083 [0.053] N=1,340	0.072 [0.058] N=1,128	0.091 [0.075] N=872
Overweight including obesity, BMIZ above +1	0.073 [0.062] N=1,694	**0.120* [0.069] N=1,548**	0.095 [0.078] N=1,340	0.109 [0.093] N=1,128	0.150 [0.126] N=872
Obesity, BMIZ above +2	0.011 [0.044] N=1,694	0.034 [0.047] N=1,548	-0.010 [0.054] N=1,340	0.020 [0.065] N=1,128	-0.025 [0.080] N=872

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.068	-0.119 [0.127] N=2,025	-0.202 [0.136] N=1,878	-0.198 [0.139] N=1,671	-0.154 [0.147] N=1,459	-0.153 [0.157] N=1,200
Weight-for-age z score, through 120 months	0.339	0.699 [1.266] N=43	1.126 [1.496] N=40	0.978 [1.400] N=38	2.075 [2.020] N=33	3.197 [1.941] N=29
BMI-for-age z score	0.425	-0.041 [0.149] N=2,021	0.018 [0.155] N=1,875	-0.115 [0.168] N=1,667	-0.080 [0.181] N=1,455	-0.116 [0.196] N=1,199
Stunted, HAZ below -2	0.099 P=200	0.005 [0.027] N=2,025	0.004 [0.028] N=1,878	0.010 [0.028] N=1,671	0.018 [0.030] N=1,459	0.021 [0.033] N=1,200
Underweight, WAZ below -2, through 120 months	0.028 P=1	-0.274 [0.215] N=43	-0.308 [0.237] N=40	-0.297 [0.223] N=38	-0.385 [0.276] N=33	-0.294 [0.212] N=29
Thinness, BMIZ below -2	0.059 P=115	0.004 [0.024] N=2,021	-0.002 [0.026] N=1,875	0.014 [0.028] N=1,667	0.000 [0.028] N=1,455	-0.004 [0.030] N=1,199
Overweight including obesity, BMIZ above +1	0.378 P=773	0.044 [0.048] N=2,021	0.064 [0.051] N=1,875	0.040 [0.055] N=1,667	0.042 [0.060] N=1,455	0.041 [0.069] N=1,199
Obesity, BMIZ above +2	0.126 P=258	-0.047 [0.032] N=2,021	-0.041 [0.033] N=1,875	**-0.074** [0.035] N=1,667**	**-0.072* [0.038] N=1,455**	**-0.102** [0.041] N=1,199**

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.229 [0.195] N=1,697	**-0.397* [0.216] N=1,550**	**-0.478** [0.229] N=1,343**	-0.433 [0.272] N=1,131	**-0.600* [0.322] N=872**
Weight-for-age z score, through 120 months	-0.193 [1.905] N=34	1.276 [2.541] N=31	0.496 [2.536] N=29	6.979 [4.703] N=24	6.949 [5.385] N=20
BMI-for-age z score	-0.091 [0.222] N=1,694	0.032 [0.244] N=1,548	-0.208 [0.279] N=1,340	-0.155 [0.329] N=1,128	-0.244 [0.404] N=872
Stunted, HAZ below -2	0.022 [0.044] N=1,697	0.017 [0.050] N=1,550	0.032 [0.049] N=1,343	0.047 [0.059] N=1,131	0.085 [0.071] N=872
Underweight, WAZ below -2, through 120 months	-0.303 [0.246] N=34	-0.391 [0.324] N=31	-0.315 [0.293] N=29	-0.580 [0.491] N=24	-0.360 [0.535] N=20
Thinness, BMIZ below -2	0.050 [0.041] N=1,694	0.045 [0.045] N=1,548	0.083 [0.053] N=1,340	0.070 [0.058] N=1,128	0.090 [0.075] N=872
Overweight including obesity, BMIZ above +1	0.074 [0.062] N=1,694	**0.120* [0.068] N=1,548**	0.094 [0.078] N=1,340	0.114 [0.093] N=1,128	0.161 [0.123] N=872
Obesity, BMIZ above +2	0.011 [0.044] N=1,694	0.034 [0.048] N=1,548	-0.012 [0.054] N=1,340	0.021 [0.065] N=1,128	-0.025 [0.080] N=872

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.068	-0.107 [0.124] N=2,025	-0.189 [0.132] N=1,878	-0.186 [0.136] N=1,671	-0.131 [0.144] N=1,459	-0.153 [0.155] N=1,200
Weight-for-age z score, through 120 months	0.339	1.106 [1.227] N=43	1.224 [1.276] N=40	0.982 [1.120] N=38	1.855 [1.319] N=33	**2.635** [1.075] N=29**
BMI-for-age z score	0.425	-0.040 [0.148] N=2,021	0.020 [0.154] N=1,875	-0.109 [0.167] N=1,667	-0.070 [0.179] N=1,455	-0.112 [0.195] N=1,199
Stunted, HAZ below -2	0.099 P=200	0.001 [0.027] N=2,025	0.001 [0.028] N=1,878	0.006 [0.028] N=1,671	0.013 [0.031] N=1,459	0.019 [0.033] N=1,200
Underweight, WAZ below -2, through 120 months	0.028 P=1	-0.271 [0.204] N=43	-0.271 [0.206] N=40	-0.243 [0.177] N=38	-0.267 [0.193] N=33	-0.190 [0.147] N=29
Thinness, BMIZ below -2	0.059 P=115	0.004 [0.024] N=2,021	-0.002 [0.025] N=1,875	0.013 [0.028] N=1,667	-0.001 [0.027] N=1,455	-0.004 [0.029] N=1,199
Overweight including obesity, BMIZ above +1	0.378 P=773	0.042 [0.048] N=2,021	0.063 [0.051] N=1,875	0.040 [0.054] N=1,667	0.041 [0.059] N=1,455	0.041 [0.068] N=1,199
Obesity, BMIZ above +2	0.126 P=258	-0.048 [0.032] N=2,021	-0.041 [0.033] N=1,875	**-0.073** [0.035] N=1,667**	**-0.070* [0.039] N=1,455**	**-0.101** [0.041] N=1,199**

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.204 [0.194] N=1,697	**-0.372* [0.215] N=1,550**	**-0.451* [0.229] N=1,343**	-0.382 [0.272] N=1,131	**-0.588* [0.328] N=872**
Weight-for-age z score, through 120 months	0.428 [2.225] N=34	0.879 [2.460] N=31	-0.300 [1.953] N=29	1.659 [3.250] N=24	6.117 [5.590] N=20
BMI-for-age z score	-0.071 [0.219] N=1,694	0.057 [0.240] N=1,548	-0.177 [0.272] N=1,340	-0.106 [0.323] N=1,128	-0.206 [0.393] N=872
Stunted, HAZ below -2	0.016 [0.045] N=1,697	0.012 [0.050] N=1,550	0.027 [0.050] N=1,343	0.038 [0.061] N=1,131	0.082 [0.073] N=872
Underweight, WAZ below -2, through 120 months	-0.373 [0.294] N=34	-0.424 [0.360] N=31	-0.294 [0.290] N=29	-0.104 [0.305] N=24	0.104 [0.334] N=20
Thinness, BMIZ below -2	0.049 [0.041] N=1,694	0.043 [0.045] N=1,548	0.082 [0.053] N=1,340	0.067 [0.057] N=1,128	0.088 [0.073] N=872
Overweight including obesity, BMIZ above +1	0.075 [0.062] N=1,694	**0.124* [0.068] N=1,548**	0.100 [0.078] N=1,340	0.116 [0.093] N=1,128	0.163 [0.126] N=872
Obesity, BMIZ above +2	0.013 [0.044] N=1,694	0.037 [0.047] N=1,548	-0.009 [0.054] N=1,340	0.026 [0.066] N=1,128	-0.024 [0.081] N=872

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.085	-0.113 [0.133] N=1,914	-0.201 [0.142] N=1,777	-0.201 [0.147] N=1,576	-0.182 [0.159] N=1,375	-0.220 [0.171] N=1,135
Weight-for-age z score, through 120 months	0.318	0.125 [1.344] N=41	0.823 [1.416] N=38	0.368 [1.234] N=36	0.514 [1.282] N=31	0.817 [1.585] N=28
BMI-for-age z score	0.430	-0.051 [0.157] N=1,911	0.020 [0.163] N=1,775	-0.096 [0.178] N=1,573	-0.062 [0.193] N=1,372	-0.112 [0.209] N=1,134
Stunted, HAZ below -2	0.101 P=195	-0.004 [0.028] N=1,914	-0.005 [0.030] N=1,777	0.002 [0.030] N=1,576	0.017 [0.032] N=1,375	0.024 [0.034] N=1,135
Underweight, WAZ below -2, through 120 months	0.031 P=1	-0.240 [0.245] N=41	-0.244 [0.244] N=38	-0.200 [0.212] N=36	-0.166 [0.215] N=31	-0.041 [0.179] N=28
Thinness, BMIZ below -2	0.058 P=107	-0.003 [0.026] N=1,911	-0.010 [0.027] N=1,775	0.003 [0.030] N=1,573	-0.005 [0.029] N=1,372	-0.006 [0.031] N=1,134
Overweight including obesity, BMIZ above +1	0.379 P=732	0.026 [0.050] N=1,911	0.050 [0.053] N=1,775	0.027 [0.058] N=1,573	0.032 [0.062] N=1,372	0.024 [0.071] N=1,134
Obesity, BMIZ above +2	0.127 P=248	-0.043 [0.032] N=1,911	-0.034 [0.034] N=1,775	-0.061 [0.037] N=1,573	-0.053 [0.041] N=1,372	**-0.084* [0.044] N=1,134**

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.201 [0.198] N=1,602	**-0.382* [0.222] N=1,465**	**-0.461* [0.239] N=1,264**	**-0.476* [0.286] N=1,063**	**-0.742** [0.340] N=823**
Weight-for-age z score, through 120 months	-0.357 [2.006] N=32	1.809 [2.494] N=29	0.997 [2.623] N=27	1.895 [4.322] N=22	1.543 [6.956] N=19
BMI-for-age z score	-0.121 [0.232] N=1,600	0.020 [0.254] N=1,464	-0.190 [0.290] N=1,262	-0.156 [0.348] N=1,061	-0.328 [0.422] N=823
Stunted, HAZ below -2	0.009 [0.046] N=1,602	0.004 [0.052] N=1,465	0.022 [0.051] N=1,264	0.051 [0.062] N=1,063	0.104 [0.074] N=823
Underweight, WAZ below -2, through 120 months	-0.302 [0.328] N=32	-0.381 [0.339] N=29	-0.186 [0.304] N=27	0.232 [0.405] N=22	0.630 [0.699] N=19
Thinness, BMIZ below -2	0.049 [0.043] N=1,600	0.041 [0.047] N=1,464	0.076 [0.055] N=1,262	0.075 [0.060] N=1,061	0.112 [0.075] N=823
Overweight including obesity, BMIZ above +1	0.058 [0.067] N=1,600	0.111 [0.073] N=1,464	0.089 [0.084] N=1,262	0.115 [0.101] N=1,061	0.134 [0.136] N=823
Obesity, BMIZ above +2	0.004 [0.045] N=1,600	0.031 [0.049] N=1,464	-0.005 [0.056] N=1,262	0.034 [0.069] N=1,061	-0.017 [0.085] N=823

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.085	-0.107 [0.131] N=1,914	-0.197 [0.141] N=1,777	-0.197 [0.147] N=1,576	-0.171 [0.158] N=1,375	-0.203 [0.167] N=1,135
Weight-for-age z score, through 120 months	0.318	0.159 [1.251] N=41	0.717 [1.314] N=38	0.529 [1.233] N=36	0.810 [1.272] N=31	1.258 [1.467] N=28
BMI-for-age z score	0.430	-0.054 [0.154] N=1,911	0.019 [0.159] N=1,775	-0.093 [0.173] N=1,573	-0.071 [0.186] N=1,372	-0.121 [0.201] N=1,134
Stunted, HAZ below -2	0.101 P=195	-0.006 [0.028] N=1,914	-0.006 [0.030] N=1,777	0.000 [0.030] N=1,576	0.014 [0.033] N=1,375	0.020 [0.035] N=1,135
Underweight, WAZ below -2, through 120 months	0.031 P=1	-0.222 [0.219] N=41	-0.225 [0.214] N=38	-0.208 [0.198] N=36	-0.178 [0.206] N=31	-0.108 [0.180] N=28
Thinness, BMIZ below -2	0.058 P=107	-0.003 [0.025] N=1,911	-0.011 [0.026] N=1,775	0.002 [0.029] N=1,573	-0.006 [0.029] N=1,372	-0.007 [0.031] N=1,134
Overweight including obesity, BMIZ above +1	0.379 P=732	0.025 [0.049] N=1,911	0.049 [0.052] N=1,775	0.028 [0.056] N=1,573	0.031 [0.061] N=1,372	0.025 [0.070] N=1,134
Obesity, BMIZ above +2	0.127 P=248	-0.046 [0.032] N=1,911	-0.037 [0.034] N=1,775	**-0.063* [0.037] N=1,573**	-0.057 [0.041] N=1,372	**-0.088** [0.043] N=1,134**

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.210 [0.198] N=1,602	**-0.395* [0.221] N=1,465**	**-0.483** [0.238] N=1,264**	**-0.490* [0.286] N=1,063**	**-0.754** [0.340] N=823**
Weight-for-age z score, through 120 months	-0.598 [1.920] N=32	0.643 [2.195] N=29	-0.324 [2.061] N=27	1.815 [3.781] N=22	1.167 [6.520] N=19
BMI-for-age z score	-0.124 [0.231] N=1,600	0.020 [0.251] N=1,464	-0.175 [0.286] N=1,262	-0.139 [0.342] N=1,061	-0.306 [0.417] N=823
Stunted, HAZ below -2	0.010 [0.046] N=1,602	0.006 [0.052] N=1,465	0.026 [0.052] N=1,264	0.054 [0.062] N=1,063	0.108 [0.073] N=823
Underweight, WAZ below -2, through 120 months	-0.292 [0.308] N=32	-0.295 [0.274] N=29	-0.197 [0.232] N=27	0.249 [0.341] N=22	0.692 [0.691] N=19
Thinness, BMIZ below -2	0.048 [0.043] N=1,600	0.040 [0.047] N=1,464	0.074 [0.055] N=1,262	0.074 [0.059] N=1,061	0.111 [0.075] N=823
Overweight including obesity, BMIZ above +1	0.057 [0.066] N=1,600	0.111 [0.072] N=1,464	0.094 [0.082] N=1,262	0.120 [0.098] N=1,061	0.142 [0.133] N=823
Obesity, BMIZ above +2	0.003 [0.045] N=1,600	0.030 [0.049] N=1,464	-0.004 [0.056] N=1,262	0.034 [0.068] N=1,061	-0.016 [0.083] N=823

Notes: Cells report the reform coefficient, clustered standard error in brackets, and N; bold stars indicate *** p<0.01, ** p<0.05, * p<0.10. Five columns use 8- to 4-year parental birth-month bandwidths. The 12-month donut omits months -12 through +11. All specifications include exact child-month fixed effects, split linear trends, and FAKTOR_FERT survey weights. Specification 2 adds female; specifications 3 and 4 separately add household and per-person income groups; specifications 5 and 6 separately add aid count and any aid. WHO 2007 HAZ and BMIZ are valid at months 61-228; WAZ ends at month 120. Implausible z scores are excluded. P is the positive-case count in the eight-year no-donut sample. NA indicates zero standard error. School-level and measurement selection affect interpretation.